

Study Questions: Cognition — Mental Models of the Finished Piece

1. When you read a crochet pattern and visualize the finished piece, what cognitive process are you performing in Active Inference terms?
2. How does the speed at which you decode stitch abbreviations (sc, dc, fpdc) reflect the training level of your generative model?
3. Explain why a pattern instruction like "inc evenly across" requires mental arithmetic. What is your generative model computing?
4. Describe how an experienced crocheter can look at their work-in-progress and determine which row they are on. What cognitive inference is happening?
5. How does recognizing a pattern repeat (e.g., a 4-stitch repeat) change your cognitive processing of the pattern? What predictions does recognition enable?
6. Why might a crocheter who is fluent in US terms struggle with a UK-term pattern, even though their cognitive model of stitches is the same?
7. Explain the cognitive difference between reading a stitch abbreviation you know instantly versus one you must look up. What does the lookup reveal about your model?
8. A pattern says "work 3 dc, ch 2, 3 dc in corner space." Describe the multi-step mental simulation your model performs to predict the outcome.
9. How does the construction logic of a top-down raglan differ from a set-in sleeve garment in terms of cognitive complexity?
10. Why is "reading your fabric" (identifying row number and stitch type from the physical fabric) the inverse of visualization? What makes it cognitively challenging?
11. A crocheter who designs patterns must decompose a finished object into construction steps. What cognitive process is this in Active Inference terms?

12. How does understanding the mathematical structure of a flat circle (6 increases per round for sc) reflect model-based reasoning?
13. Describe the cognitive load difference between working a row of all single crochet and a row with 5 different stitch types. Why is the complex row harder?
14. When a pattern uses a chart instead of written instructions, what different cognitive processes are engaged? Why might some crocheters prefer one over the other?
15. How does mental arithmetic for gauge conversion (e.g., pattern gauge is 16 sc/4in but yours is 18 sc/4in) represent model-based computation?
16. Explain why experienced crocheters can often predict what a stitch pattern will look like just from reading the abbreviations, without making a swatch.
17. A pattern says "reverse sc" — a stitch you have never done. Describe the cognitive process of learning a new stitch as extending your generative model.
18. How does the cognitive experience of crocheting a familiar pattern differ from crocheting an unfamiliar one? What role does model familiarity play?
19. When you mentally plan where to start decreasing for a hat crown, you are computing future states of the fabric. How many rows ahead does your model need to predict?
20. How does the ability to mentally rotate a flat pattern piece and imagine it as part of a 3D garment reflect spatial reasoning in the generative model?